

Malaysia EN/BM Cross-Language Consistency Test

**Full C1 → C0 → C2 pipeline on parallel
Economic Report EN+BM pairs, 2014–
2025**

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Framing

In this section, we seek to answer the question: Can our pipeline produce equivalent results with two equivalent inputs that are in two different languages? And more specifically, will including fiscal documents in other languages (Malay) yield results that are comparable to the ones we obtained in English during the development phase?

The C1 → C0 → C2 codebook pipeline was validated on US English documents. Malaysia has 38 BM-only documents in its ready corpus (primarily older Budget Speeches and Economic Reports). Before trusting LLM outputs on those, we need evidence that the pipeline produces **equivalent fiscal reasoning regardless of input language**.

This is a consistency test, not a validity test. We have no labels for Malaysia. EN outputs are treated as the reference because the codebooks crossed S3 gates on EN-only US data. Passing this test is necessary-but-not-sufficient for extending deployment to the **BM-only-document slice (38 of 99 ready documents)**; it does not gate the broader Malaysia deployment, and it is **not a substitute for expert agreement on the EN-side outputs** (Phase 2’s success criteria: ≥80% on C1, ≥70% on C2 against expert review).

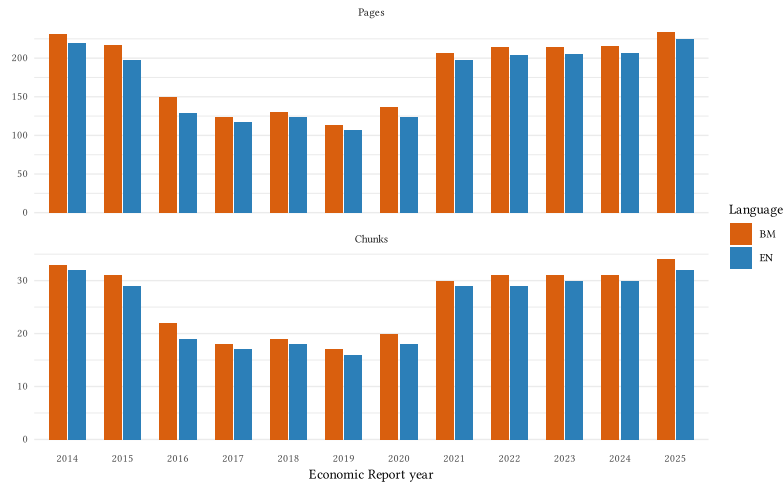
The test runs the **full pipeline independently on each language** and compares the outputs within each step of the process. It

restricts the corpus to Economic Reports with parallel EN+BM coverage, runs its own C1 $\rightarrow$ C2a chain, then aggregates measure names into acts with the **C0 act aggregator** at three scopes:

- **per-document** — aggregate within each language $\times$ year document (granular aggregation capability/limits);
- **per-language** — pool all EN documents / all BM documents (deployment-realistic; this scope feeds C2 and the headline timeline);
- **joint EN+BM** — pool both languages together (probes whether C0 can co-cluster the EN and BM surface forms of the same act).

Scope

Figure 1 shows the extent to which the two language sides cover the same document: the page count and chunk count of each Economic Report year, by language. Comparable bars within a year mean we are comparing parallel documents; a large gap flags an edition or coverage asymmetry (most visibly in 2023, discussed below).



This test only speaks to **modern, professionally-translated MoF prose** (Economic Reports 2014–2025; see the 2023 edition caveat under C1-step comparability). It does NOT speak to pre-2009 Budget Speech BM, older Economic Report BM (pre-2014), or crisis-booklet BM. Those are different vintages and registers; deployment to them requires separate evidence.

Figure 1: Pages and chunks per Economic Report year, by language. Comparable EN/BM bars within a year indicate parallel coverage.

C1-step comparability

Before aggregation, does C1 surface a comparable set of raw fiscal measures on each side? This is the step where real extraction asymmetries live (independent of how acts are later aggregated). Figure 2 counts the distinct fiscal measures C1 produced per document, by language.

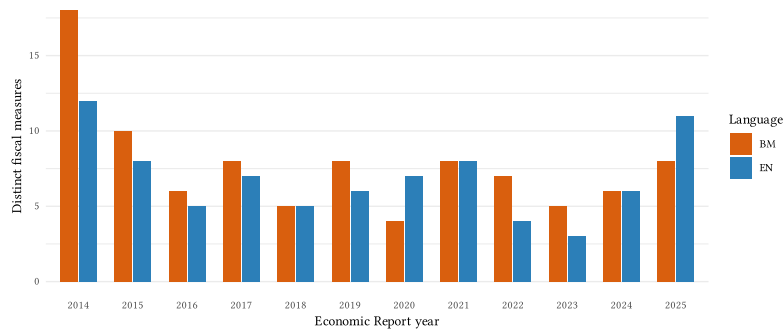


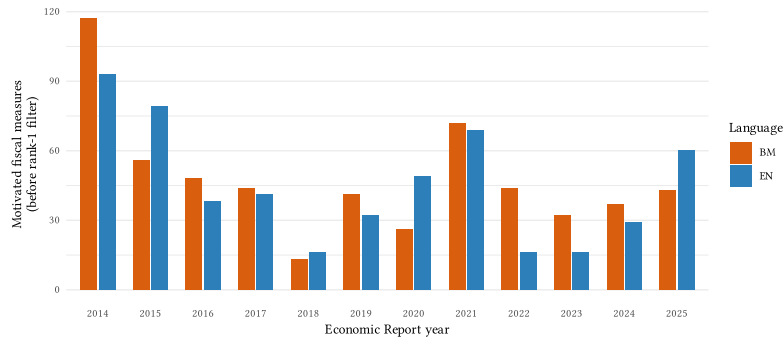
Figure 2: Distinct C1 fiscal-measure names per year, by language (pre-aggregation). A large within-year EN/BM gap is an extraction asymmetry that bounds any downstream aggregation.

Two C1-level residuals carry over from the prior iteration’s analysis and are **not** aggregation artifacts: (i) some EN/BM years contain **foreign-comparator** measures (Japan, India, Australia stimulus packages; Philippines TRAIN Act) that should be filtered upstream by country-of-enactment tagging in C1; and (ii) **2018 BM** extracted suspiciously few measures from a full Economic Report, a recall asymmetry to investigate at C1, not C0. These belong to the C1 step and bound how comparable any downstream aggregation can be.

Why the drift varies by year

The Figure 2 drift is not uniform across years; it concentrates in 2014 (the GST-announcement Budget). Two structural features of each year’s Economic Report drive it, and both peak in 2014.

First, **measure density**. The comparability count above is a post-filter count: the deployment filter keeps only the most-prominent (rank-1) motivated measure per chunk. Before that cut, C1 surfaces several measures per chunk, and 2014 carries roughly two to three times the motivated-fiscal-measure load of a typical year. The denser the report, the more measures the rank-1 cut discards, and the more room chunk-boundary differences between the parallel EN and BM documents have to change *which* measure survives on each side.



Warning

The 2023 EN and BM documents are **different editions of the 2023 budget cycle**, not parallel translations. The BM side is the Oct-2022 full *Tinjauan Ekonomi 2023*; the EN side is the Feb-2023 re-tabled-budget *Updates on Economic and Fiscal Outlook and Revenue Estimates 2023* — a condensed Outlook plus a bundled Federal Government Revenue Estimates / Treasury Memorandum block that has no BM counterpart here (or in any 2014–2022 EN file). Expect **EN-heavy C1 measure counts for 2023**; the drift on that row reflects document-edition asymmetry, not a language/reasoning inconsistency. See the 2026-06-05 provenance note in [data/manual/malaysia/README.md](#).

Figure 3: Motivated fiscal measures C1 surfaces per year by language, before the rank-1 deployment filter. 2014 carries the heaviest load of any pilot year, which is what the rank-1 cut discards most and most asymmetrically.

Second, **chunk-level classification asymmetry**. With near-identical chunk counts across the two parallel documents, BM flags a higher share of its 2014 chunks as carrying a motivated rank-1 fiscal measure than EN does. Since each such chunk contributes exactly one measure to Table 1, this EN/BM gap *is* the drift, and it is widest in 2014.

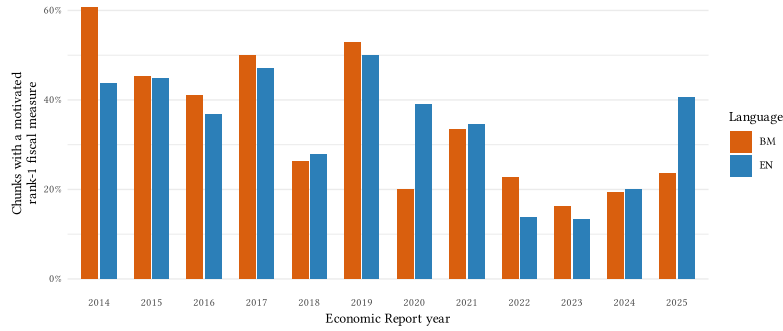


Figure 4: Share of each year's chunks whose most-prominent (rank-1) measure is a motivated fiscal measure, by language. The EN/BM gap, the per-chunk asymmetry the drift reflects, is widest in 2014.

C0 aggregation diagnostics

C0 collapses C1 measure names into act-level clusters. The three scopes test different properties of that aggregation.

Per-document: granular aggregation symmetry

Per document, Figure 5 shows the compression ratio — how many surfaced measures collapse into each distinct act — by year and language. A symmetric aggregator should compress comparably on both sides; the previous clusterer fragmented BM more than EN.

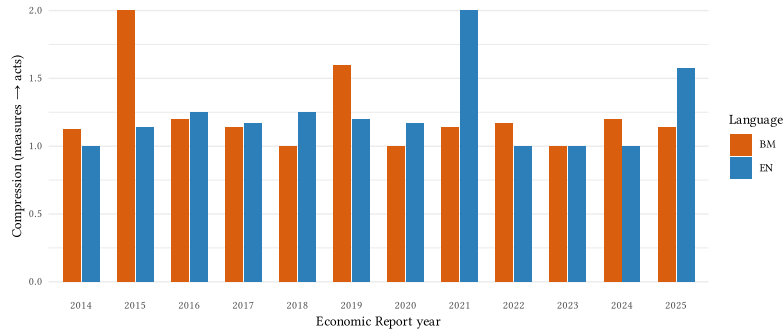


Figure 5: C0 per-document compression (measures collapsed into distinct acts) by year and language. Comparable EN/BM bars indicate symmetric aggregation.

Per-language: act inventory marginals

Pooling all documents per language, how do the two full-pipeline runs compare on act count and the exogenous/endogenous split? Identical marginals are necessary-but-not-sufficient for per-act consistency (the timeline below addresses the rest), but a large marginal gap is itself a finding.

Table 1: Per-language act inventory: counts and exogeneity split

Language	C0 acts	Exogenous	Endogenous
BM	72	53	19
EN	59	41	18

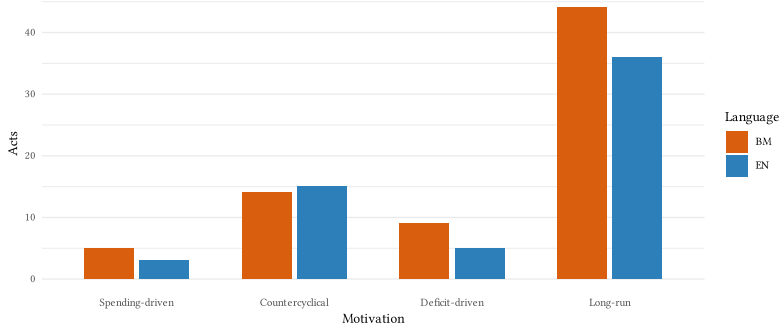


Figure 6: Motivation-label marginal distribution by language. A BM-side over-representation of a structural label (e.g. Long-run) surfaces here as a marginal shift — no act-by-act pairing required.

Joint EN+BM: cross-language merge rate

Run on the combined EN+BM pool, how often does C0 place an EN surface form and a BM surface form of the same act into one cluster? This is the direct, no-API probe of whether C0 bridges languages (e.g. *Subsidi Bahan Api* $\leftrightarrow$ *fuel subsidy*). A low merge rate means C0 treats the languages as disjoint and cross-language aggregation needs a translation step.

Table 2: Joint-pool clustering: mixed (EN+BM) vs single-language clusters

Total clusters	Mixed (EN+BM)	EN-only	BM-only	Merge rate
101	25	32	44	24.8%

When the merge rate is non-trivial, audit a few mixed clusters by eye (do the EN and BM members really name the same act?). This eyeball check is the human-verified replacement for the dropped Sonnet matcher.

Headline: final-output consistency

The pipeline runs end-to-end on each language and produces a set of acts, each carrying a C2 motivation label, sign, and exogeneity. The question is whether the two runs land on the same fiscal reasoning. Pure marginals cannot answer this (identical totals can hide compensating per-act errors), so the headline is **two timeline figures**: for each year and language, acts are counted into motiva-

tion $\times$ sign cells and drawn as a diverging stacked bar — increases grow the bar upward, decreases downward, coloured by motivation, faceted by language. The reader compares the EN and BM rows directly.

The two figures differ only in the timing source: the first dates each act by the year **extracted from its name** (language-invariant numerals; partial coverage), the second by the **source-document year** (always present).

Act-name year multiset

A quick deterministic check before the plots: do the two sides surface acts carrying the same set of name-embedded years?

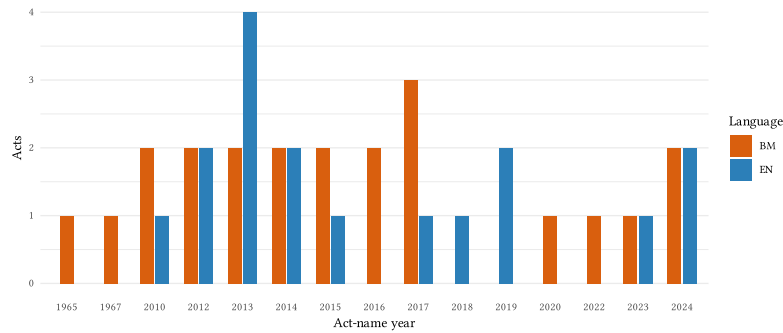


Figure 7: Act-name-year multiset by language (years parsed from act names). Comparable EN/BM bars mean the two sides surface acts dated to the same years.

Timeline by act-name year



Figure 8: Act counts by year extracted from the act name, EN vs BM. Diverging stacked bars: increases grow upward, decreases downward, no-change acts sit at the baseline; colour = motivation; faceted by language.

Timeline by source-document year



Figure 9: Act counts by modal source-document year, EN vs BM. Same encodings as above.

Interpretation

Read the evidence in pipeline order:

- **C1 step.** Any large pre-aggregation drift, foreign-comparator contamination, or the 2018 BM recall gap is a C1 issue and bounds everything downstream — it is not fixable by the aggregator.
- **C0 step.** The per-document compression figure closes the loop opened last iteration: does C0 produce **symmetric** act-level aggregation across languages where the previous clusterer fragmented BM more than EN? The joint merge rate separately tells us whether C0 bridges languages at all or treats them as disjoint vocabularies.
- **C2 / headline.** The two timelines are where cross-language consistency of the *full* pipeline is read. Agreement looks like the same years carrying the same colours and the same up/down balance on both facets; systematic colour shifts (e.g. BM reaching for a structural label) or sign flips on the same act are the disagreements worth chasing. The act-name-year multiset and label-marginal figures are the statistical companions to the visual read.

i Methodological caveat: translation, not independent source

The BM Economic Reports are MoF translations of the EN versions, not independent fiscal-policy statements. Two consequences: **correlated errors are invisible** (if Haiku misreads a passage the same way in both languages because the translation preserved the misleading phrasing, this test records agreement); and **some disagreements are translation artifacts** (the translator’s word choice, not the model’s reasoning). The motivation language R&R rely on — “the actual words used by policymakers” — only fully applies to the EN side. Catching correlated errors requires expert validation on the EN side.

Implications for deployment scope

This test, conditional on its results, speaks to a narrow population of BM documents:

- **Speaks to:** professional MoF translations of modern Economic Reports (2014+), commissioned alongside the EN versions and expected to be high-quality.
- **Does NOT speak to:** older Economic Report BM (1995-2013, 19 documents — MoF Bahasa originals, not parallel translations), all Budget Speech BM (9 documents, 2000-2008 + 2012 + 2015 — no EN parallels, different register), or crisis-booklet BM (4 documents — branded marketing register).

The pilot window now spans the full parallel EN+BM Economic Report record (2014–2025), and the consistency test auto-includes any year for which both an EN and a BM Economic Report are available, so adding a future pair requires only supplying the two documents.